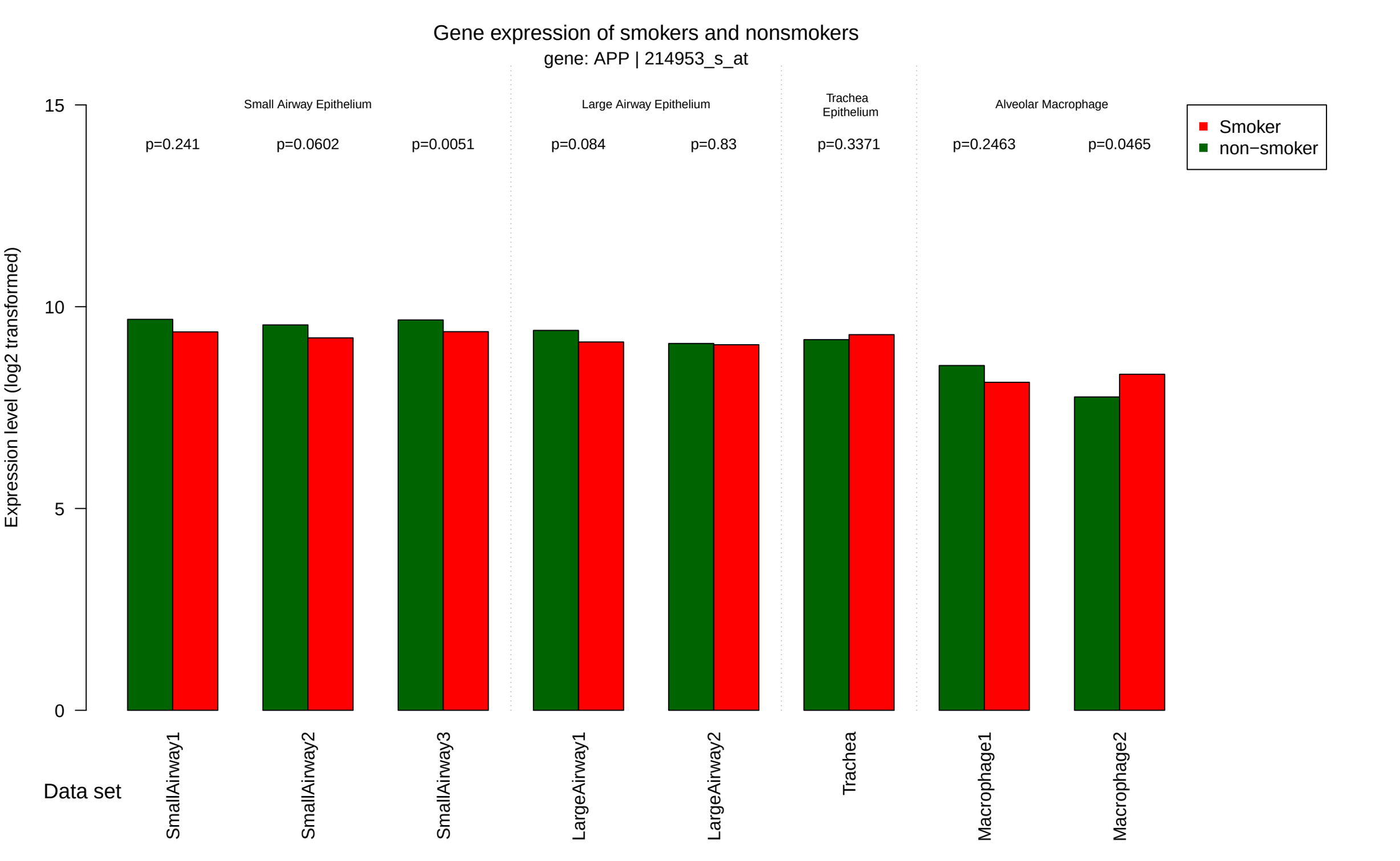
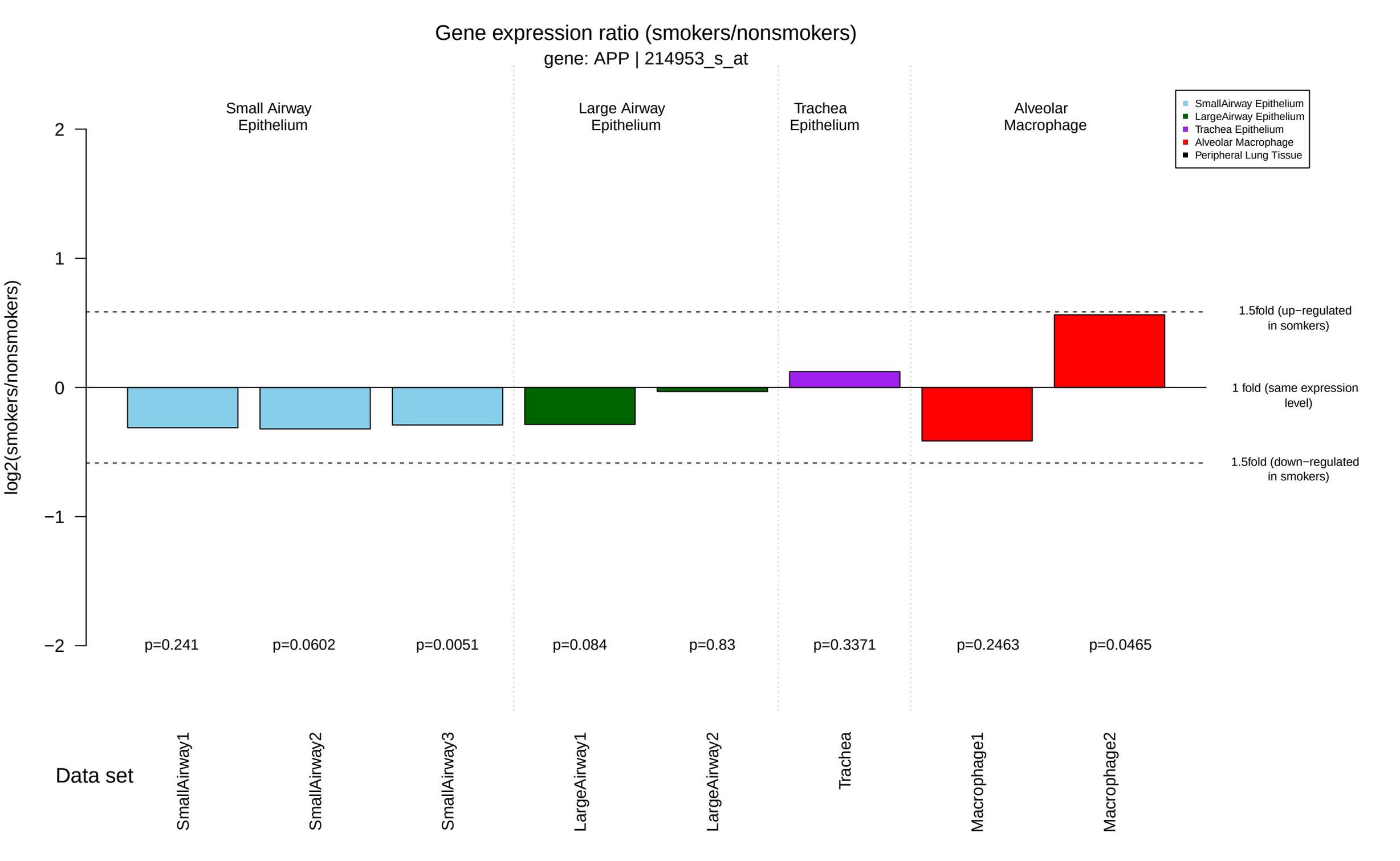
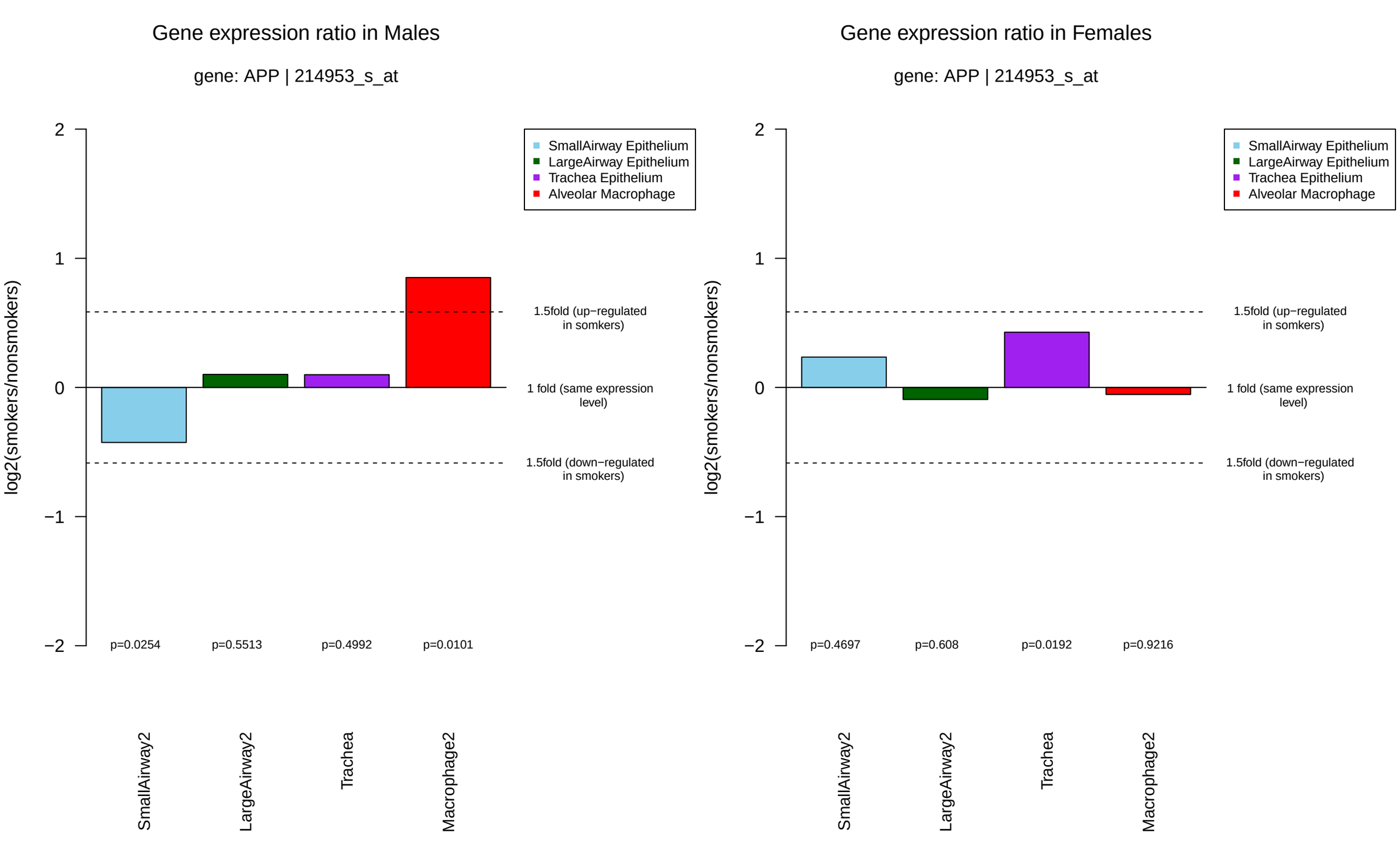
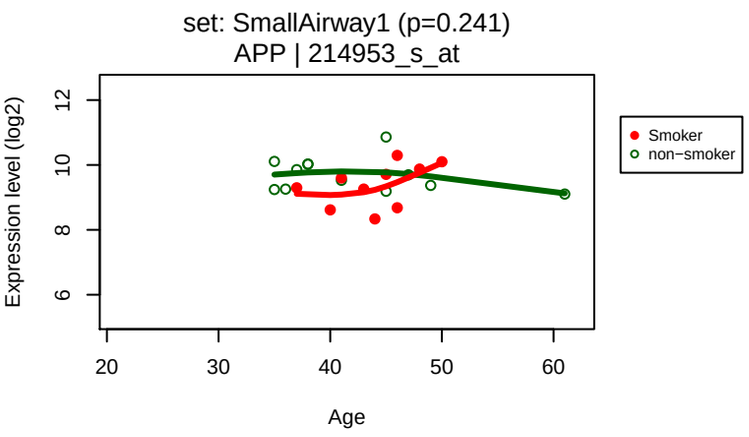
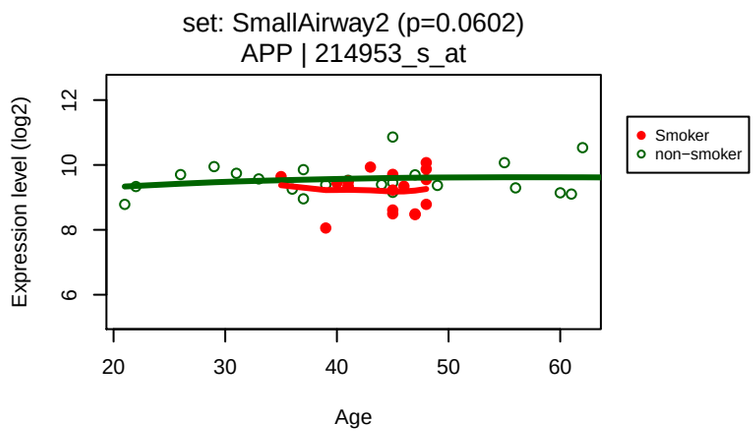
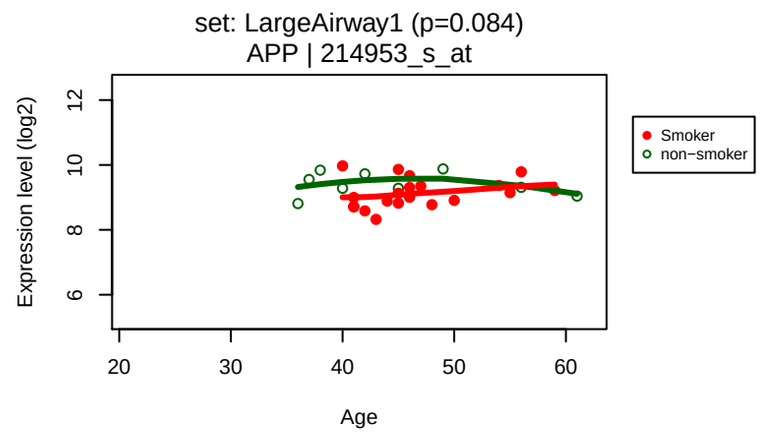

Gene expression of smokers and nonsmokers
gene: APP | 214953_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.241
p=0.0602
p=0.0051
p=0.084
p=0.83
p=0.3371
p=0.2463
p=0.0465
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

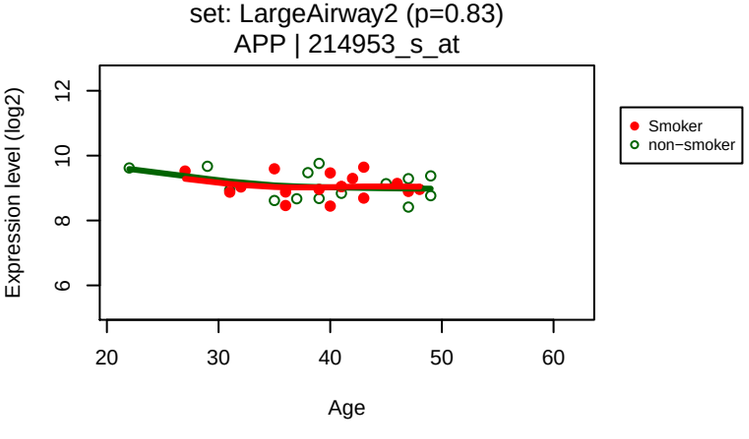

Gene expression ratio (smokers/nonsmokers)
gene: APP | 214953_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.241
p=0.0602
p=0.0051
p=0.084
p=0.83
p=0.3371
p=0.2463
p=0.0465
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

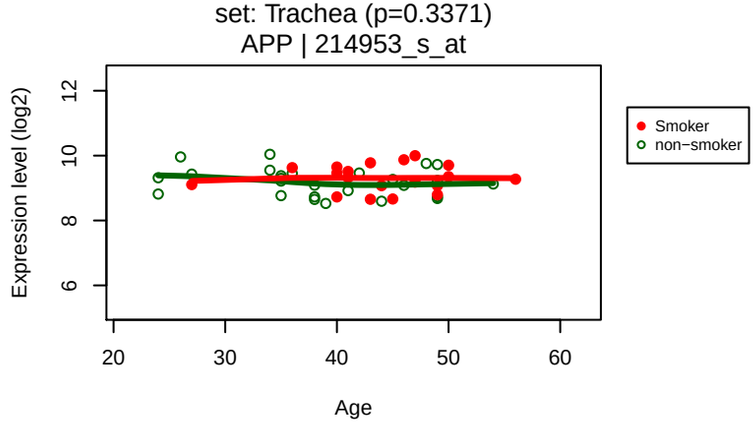

Gene expression ratio in Males
gene: APP | 214953_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0254
p=0.5513
p=0.4992
p=0.0101
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APP | 214953_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4697
p=0.608
p=0.0192
p=0.9216
SmallAirway2
LargeAirway2
Trachea
Macrophage2

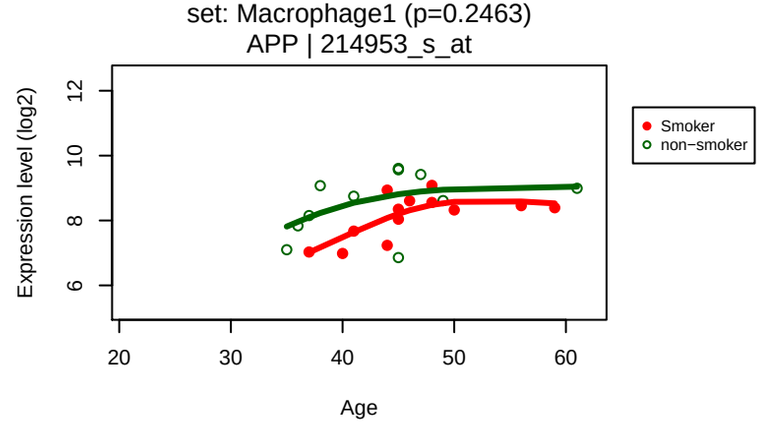

set: SmallAirway1 (p=0.241)
APP | 214953_s_at
Expression level (log2)
Age
Smoker
non−smoker

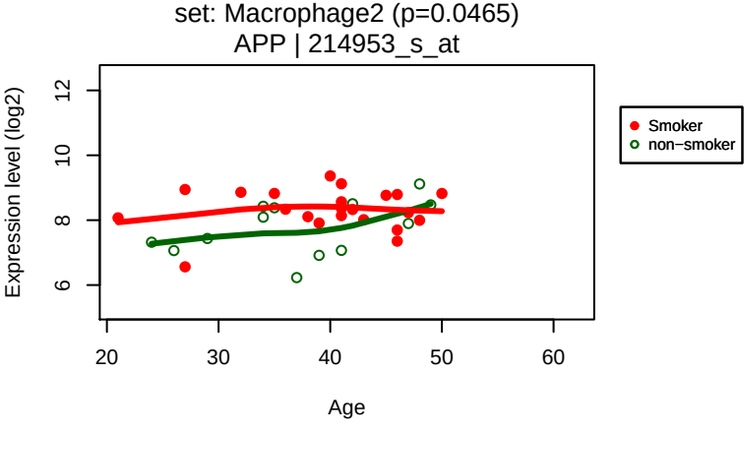

set: SmallAirway2 (p=0.0602)
APP | 214953_s_at
Expression level (log2)
Age
Smoker
non−smoker

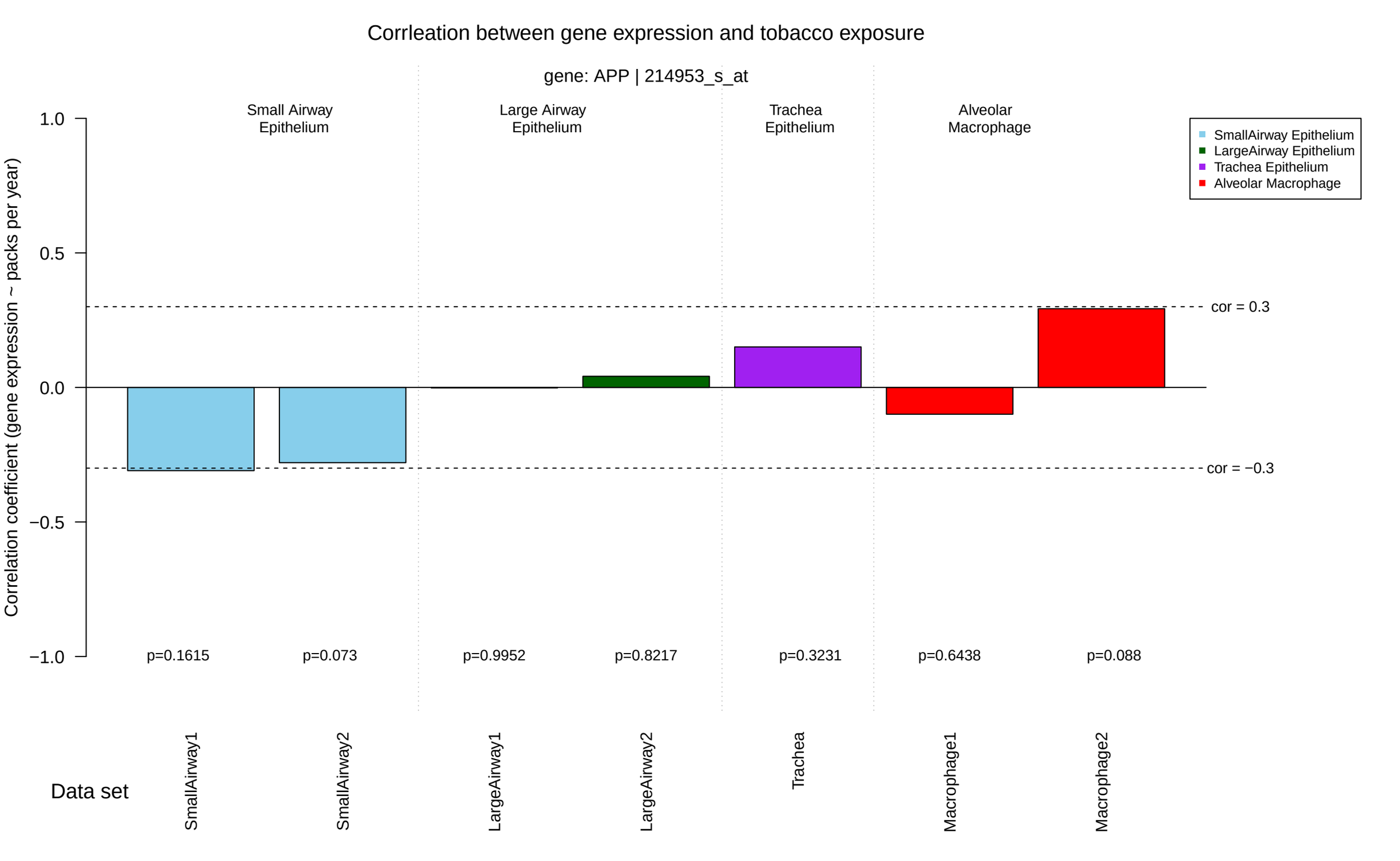

set: LargeAirway1 (p=0.084)
APP | 214953_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.83)
APP | 214953_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3371)
APP | 214953_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2463)
APP | 214953_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0465)
APP | 214953_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APP | 214953_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1615
p=0.073
p=0.9952
p=0.8217
p=0.3231
p=0.6438
p=0.088
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2